

Random Experiments, Sample Spaces and Types of Events

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Leads to:** Empirical vs theoretical probability, probability rules ([Note 331](#)); Venn diagrams and contingency tables ([Note 340](#)); Conditional probability ([Note 341](#)).

1. Overview

Key point: Probability is always measured for an event; to name an event we first need the random experiment, its trials and outcomes, and the sample space the event is taken from.

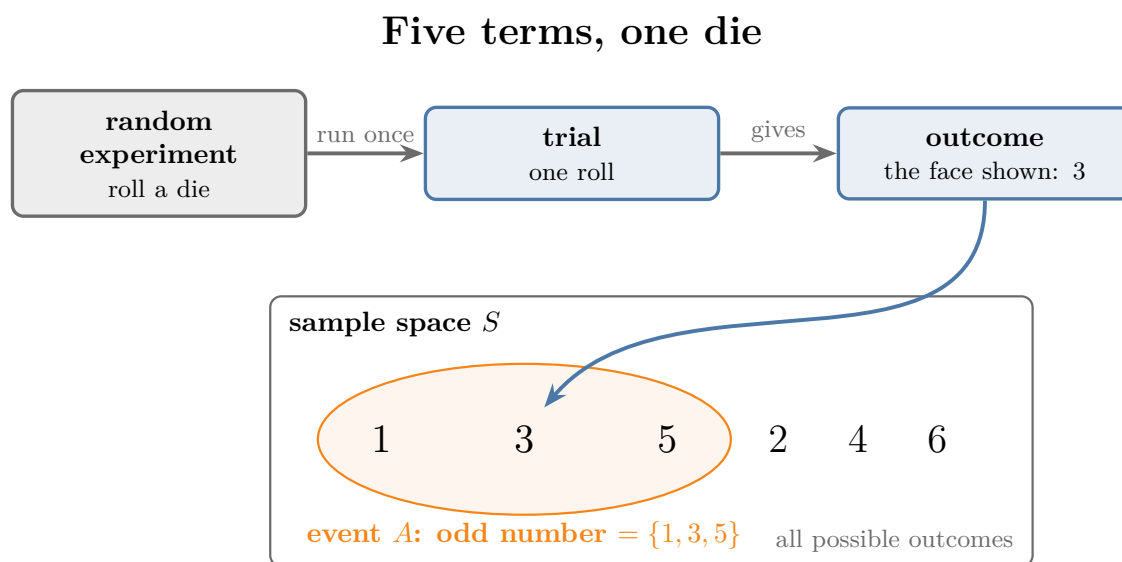


Figure 1: The five basic terms on one die roll

Figure 1 places the five basic terms of probability on a single die roll. A random experiment is run once in a trial, which gives one outcome. Every outcome lies in the sample space S , and an event is any set of outcomes we care about, here “an odd number”.

This Note defines the five terms, applies them to three experiments, and then sorts events into types. Every later probability topic, from conditional probability to Bayes’ theorem, is phrased in these words.

2. The five basic terms

Key point: Random experiment, trial, outcome, sample space and event: an experiment is run in trials, each trial gives one outcome, all possible outcomes form the sample space, and an event is a chosen part of it.

2.1 Random experiment

Key point: An experiment is random when it has more than one possible outcome and we cannot predict which one will happen.

The random experiment was introduced in the [random variables and distributions Note](#) as an experiment whose outcome is random. Stated precisely, an experiment is a random experiment when it meets **both** conditions:

1. It has **more than one** possible outcome.
2. The outcome **cannot be predicted** in advance.

Tossing a coin meets both: it can land heads or tails, and before the toss nobody can say which. Rolling a die and drawing a card from a pack are random experiments for the same reasons.

Extra: Each condition alone is not enough. A coin with heads on both sides fails the first condition: there is only one outcome. Dropping a stone fails the second: it has a known result, it falls.

2.2 Trial

Key point: A trial is one run of the random experiment.

A **trial** is a single execution of a random experiment. Tossing the coin once is one trial; tossing it again is a second trial.

Every trial produces exactly one outcome. Repeating an experiment means running more trials of it.

2.3 Outcome

Key point: An outcome is the single result of one trial.

An **outcome** is one possible result of a trial. If the first toss lands heads, heads is the outcome of that trial; if the second lands tails, tails is the outcome of the second.

2.4 Sample space

Key point: The sample space is the set of all possible outcomes of the experiment; whatever happens in a trial is always one of its members.

The sample space is defined in the [conditional probability Note](#) as the set of all possible outcomes of an experiment. It is written in curly brackets, usually named S (some books use Ω):

- Toss a coin: $S = \{H, T\}$.
- Roll a die: $S = \{1, 2, 3, 4, 5, 6\}$.

One random experiment has one sample space. Every trial's outcome comes from inside it, never from outside.

2.5 Event

Key point: An event is a set of outcomes we want the probability of: a subset of the sample space, with one outcome or several.

An event is a set of outcomes (see the [conditional probability Note](#)): a specific set of outcomes of a random experiment. In set language it is a **subset** of the sample space: every outcome in the event also belongs to S .

Events are what probability is computed for. Two examples:

- Toss a coin; event “getting a head” = $\{H\}$. One outcome.
- Roll a die; event “getting a number less than 3” = $\{1, 2\}$. Two outcomes.

Events are named with capital letters. One experiment can have many events, all drawn from the same sample space. On one die roll:

- $A = \text{“getting an odd number”} = \{1, 3, 5\}$
- $B = \text{“getting an even number”} = \{2, 4, 6\}$

The event is chosen by the question we ask, not by the experiment. Figure 1 marks A inside the sample space of the die.

3. The terms on three experiments

Key point: Writing out all five terms for an experiment removes the confusion between them; the experiment and sample space stay fixed, while the outcome changes with each trial and the event changes with each question.

The five terms are easy to read and easy to mix up. Laying them out side by side for three experiments shows what stays fixed and what changes.

3.1 Rolling a die

Key point: One roll gives one outcome from $\{1, \dots, 6\}$; the event can be one face or several.

Term	Rolling a die
Random experiment	Rolling a die
Trial	Rolling the die once
Outcome	The face shown, for example 3
Sample space	$\{1, 2, 3, 4, 5, 6\}$
Event	“Getting a 3” = $\{3\}$; “a number greater than 4” = $\{5, 6\}$; “an odd number” = $\{1, 3, 5\}$

Each trial can give a different outcome (3, then 4, then 5), but every trial gives exactly one.

3.2 Tossing a coin twice

Key point: When the experiment is “toss twice”, one trial is two tosses, and each outcome is a pair such as HT .

Here the experiment is the whole double toss, not one toss. One trial means tossing the coin twice, and its outcome records both results in order.

Term	Tossing a coin twice
Random experiment	Tossing a coin twice
Trial	One double toss
Outcome	For example HT : heads first, tails second
Sample space	$\{HH, HT, TH, TT\}$
Event	“Two heads” = $\{HH\}$; “at least one head” = $\{HH, HT, TH\}$

However many times we repeat the double toss, each result is one of these four pairs. Figure 2 draws the sample space with the two events as subsets.

Toss a coin twice: one sample space, many events

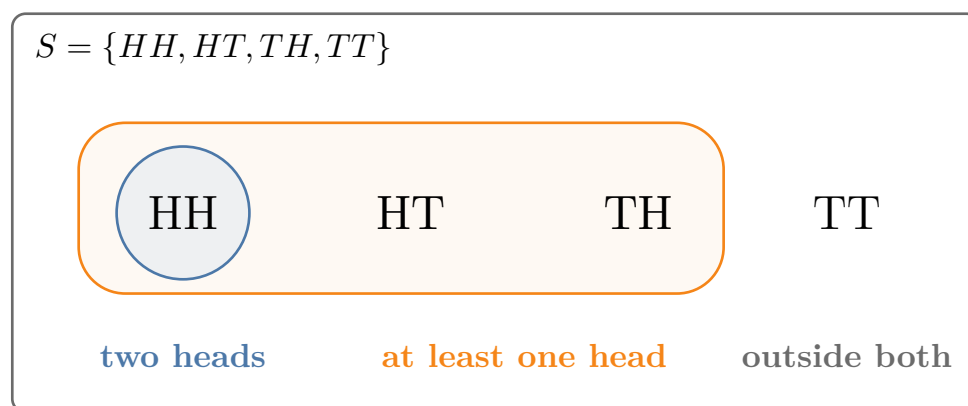


Figure 2: Tossing a coin twice: the sample space and two events

The pairs HT and TH are different outcomes: the order of the tosses is recorded. “Two heads” sits inside “at least one head”, because every outcome with two heads also has at least one.

3.3 Drawing a Titanic passenger

Key point: Picking a random row of a dataset is a random experiment too; its sample space is the set of values the chosen column can take.

The Titanic dataset (used in the [frequency tables Note](#)) lists 891 passengers. The column `Pclass` is the ticket class: 1, 2 or 3. Drawing one passenger at random and reading their class is a random experiment.

Term	Drawing a Titanic passenger
Random experiment	Drawing one passenger at random and reading <code>Pclass</code>
Trial	One draw
Outcome	For example class 1
Sample space	$\{1, 2, 3\}$
Event	“The passenger is in class 2” = $\{2\}$; “not in first class” = $\{2, 3\}$

This is how probability meets machine learning: each row of a dataset is the outcome of a trial, and a column’s possible values form a sample space. The probabilities of these events, from the counts in the data, are worked out in the [empirical and theoretical probability Note](#).

4. Types of events

Key point: Events are classified by how many outcomes they hold (simple, compound), by how two of them relate (independent, dependent, mutually exclusive, exhaustive), and by the two extremes (impossible, sure).

Since probability is always calculated for events, their types come up constantly. Figure 3 shows most of them on one die roll.

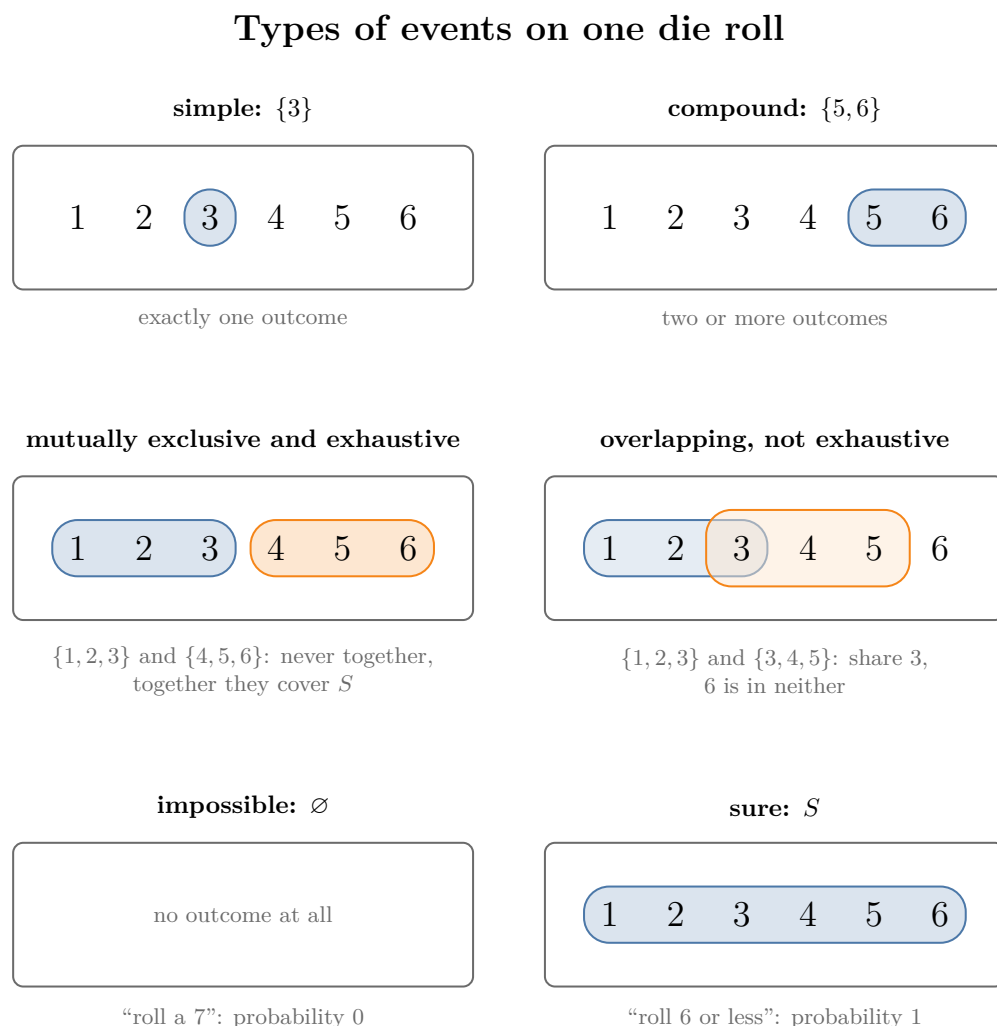


Figure 3: Types of events on one die roll

4.1 Simple events

Key point: A simple event holds exactly one outcome.

A **simple event** (also called an **elementary event**) consists of exactly one outcome. On a die, “getting a 3” = $\{3\}$ is simple (Figure 3, top left).

“Getting an odd number” is not simple: it holds three outcomes, 1, 3 and 5.

4.2 Compound events

Key point: A compound event holds two or more outcomes, so it is made of two or more simple events.

A **compound event** consists of two or more simple events. “Rolling a number greater than 4” = $\{5, 6\}$ is compound (Figure 3, top right): it is made of the simple events $\{5\}$ and $\{6\}$.

Events joined by “or” are usually compound. “Rolling a number less than 2 or greater than 3” = $\{1\} \cup \{4, 5, 6\} = \{1, 4, 5, 6\}$, where $\cup$ (union, defined in the [mutually exclusive events Note](#)) collects the outcomes of both parts.

4.3 Independent events

Key point: Two events are independent when one happening does not change the probability of the other.

Independent events are taught in the [independent events Note](#): one event happening does not change the probability of the other. Flipping a coin and rolling a die together is the standard case: the coin’s result has no effect on the die’s.

4.4 Dependent events

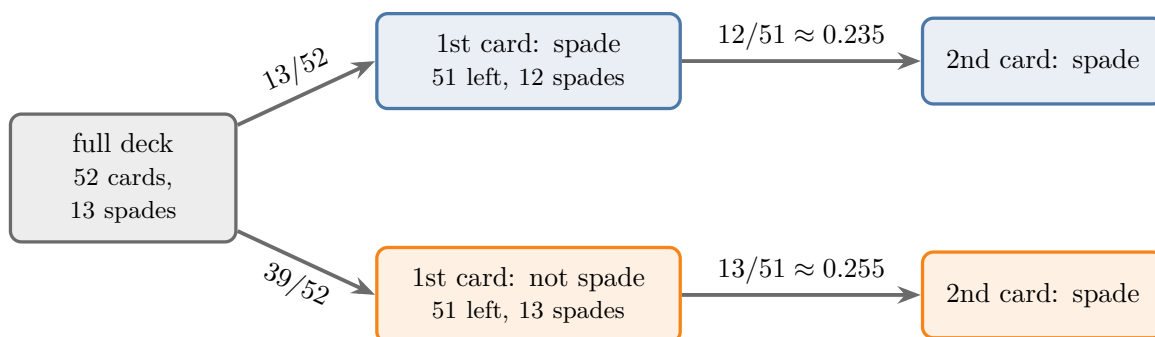
Key point: Two events are dependent when one happening changes the probability of the other, as when cards are drawn without being put back.

Two events are **dependent** when the occurrence of one **does** affect the probability of the other. They are the opposite of independent events.

Drawing two cards from a pack **without replacement** (the first card is not put back) is the standard example. A pack has 52 cards, 13 of them spades:

- First card: $P(\text{spade}) = 13/52 = 0.25$.
- Second card, if the first was a spade: 51 cards are left and 12 of them are spades, so $12/51 \approx 0.235$.
- Second card, if the first was not a spade: 13 spades are left among 51, so $13/51 \approx 0.255$.

Two cards without replacement



the second probability depends on the first card: the draws are dependent

Figure 4: Drawing two cards without replacement: the first draw changes the second

Figure 4 shows the two branches. The chance of a spade on the second draw depends on what the first draw took out, so the two draws are dependent. With replacement (the first card goes back and the pack is shuffled), the second draw is again $13/52$ whatever happened first, and the draws are independent.

Extra: The probability “spade second, given spade first” is a conditional probability, written $P(\text{2nd spade} \mid \text{1st spade}) = 12/51$; see the [conditional probability Note](#). Before we know the first card, the second card is still a spade with probability $\frac{13}{52} \cdot \frac{12}{51} + \frac{39}{52} \cdot \frac{13}{51} = \frac{1}{4}$, the same as the first. Dependence shows only once the first card is known.

4.5 Mutually exclusive events

Key point: Two events are mutually exclusive when they cannot both happen in the same trial.

Mutually exclusive events are taught in the [mutually exclusive events Note](#): they cannot happen together in one trial, so they share no outcome. On a die, $\{1, 2, 3\}$ and $\{4, 5, 6\}$ are mutually exclusive (Figure 3, middle left).

$\{1, 2, 3\}$ and $\{3, 4, 5\}$ are not (Figure 3, middle right): a roll of 3 belongs to both.

4.6 Exhaustive events

Key point: A set of events is exhaustive when together they cover the whole sample space, so at least one of them must happen in every trial.

A set of events is **exhaustive** if at least one of them must occur whenever the experiment is performed. In set terms, their outcomes together make up the whole sample space.

On a die, “odd” = $\{1, 3, 5\}$ and “even” = $\{2, 4, 6\}$ are exhaustive: every roll is odd or even. $\{1, 2, 3\}$ and $\{3, 4, 5\}$ are not exhaustive, because a roll of 6 lies in neither (Figure 3, middle right).

Exhaustive and mutually exclusive are separate properties. A set of events can have both, either one or neither:

Events on a die	Mutually exclusive?	Exhaustive?
$\{1, 2, 3\}, \{4, 5, 6\}$	yes	yes
$\{1, 2\}, \{5, 6\}$	yes	no (3 and 4 missing)
$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}$	no (share 3, 4)	yes
$\{1, 2, 3\}, \{3, 4, 5\}$	no (share 3)	no (6 missing)

Extra: Events that are both mutually exclusive and exhaustive split the sample space into separate pieces with no gaps and no overlaps. Such a set is called a **partition** of the sample space: in every trial **exactly one** of them happens. Spam and not spam, or the classes of a classifier, form a partition. The **law of total probability** needs exactly this: if $B_1, B_2, \dots$ partition the sample space, then $P(A) = \sum_i P(A | B_i) P(B_i)$ (Orloff and Bloom, MIT 18.05 *Introduction to Probability and Statistics*, Class 3 notes, Spring 2022, section “Law of total probability”). That is why the Bayes’ theorem Notes can split a probability into one term per class. The card calculation of section 4.4 is one case: “spade first” and “not spade first” partition the first draw, and $\frac{13}{52} \cdot \frac{12}{51} + \frac{39}{52} \cdot \frac{13}{51} = \frac{1}{4}$.

4.7 Impossible and sure events

Key point: The impossible event contains no outcome and has probability 0; the sure event is the whole sample space and has probability 1.

The two extreme events:

- An **impossible event** contains no outcome of the sample space. “Rolling a 7” with one die can never happen. It is the empty set, written $\emptyset$, and its probability is 0.
- A **sure event** (or **certain event**) contains every outcome. “Rolling a number from 1 to 6” always happens. It is the whole sample space S , and its probability is 1.

All other events lie between these two, with probabilities between 0 and 1 (Figure 3, bottom).

5. Summary

Term	Meaning	Die example
Random experiment	More than one outcome, none predictable	Rolling a die
Trial	One run of the experiment	One roll
Outcome	The result of one trial	3
Sample space	All possible outcomes	$\{1, 2, 3, 4, 5, 6\}$
Event	A subset of the sample space	Odd = $\{1, 3, 5\}$

Type of event	Test	Die example
Simple	Exactly one outcome	$\{3\}$
Compound	Two or more outcomes	$\{5, 6\}$
Independent	One does not change the other's probability	Coin and die together
Dependent	One changes the other's probability	Two cards without replacement
Mutually exclusive	No shared outcome	$\{1, 2, 3\}$, $\{4, 5, 6\}$
Exhaustive	Together they cover S	Odd, even
Impossible	No outcome; probability 0	Rolling a 7
Sure	All of S ; probability 1	Rolling 1 to 6

- An experiment has one sample space but can have many events: the event is set by the question.
- An outcome is one result; an event is a set of results, possibly just one.
- Mutually exclusive and exhaustive are separate properties; both together make a partition.

6. Key terms

Term	Meaning
Trial	One run of a random experiment; it gives exactly one outcome
Outcome	The single result of one trial, such as heads or a 3
Simple (elementary) event	An event with exactly one outcome
Compound event	An event with two or more outcomes
Without replacement	Drawing items without putting them back, so later draws depend on earlier ones
Exhaustive events	Events that together cover the whole sample space, so at least one always happens

Term	Meaning
Partition	Events that are mutually exclusive and exhaustive: exactly one of them happens in every trial
Impossible event	The empty event $\emptyset$; probability 0
Sure (certain) event	The whole sample space; probability 1